NATURAL LANGUAGE PROCESSING

Introduction to NLP

NLTK vs Spacy

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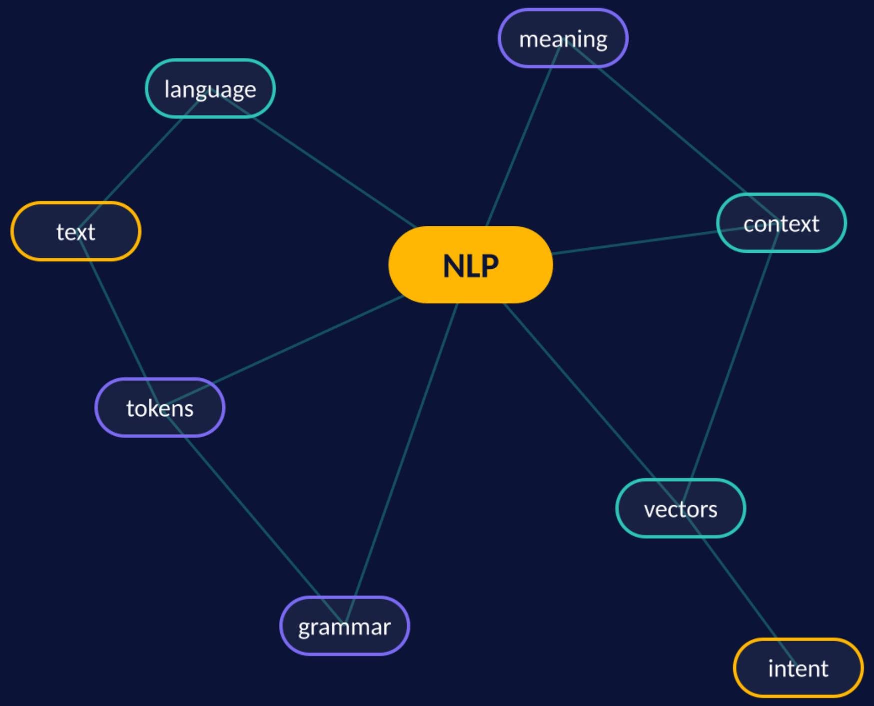
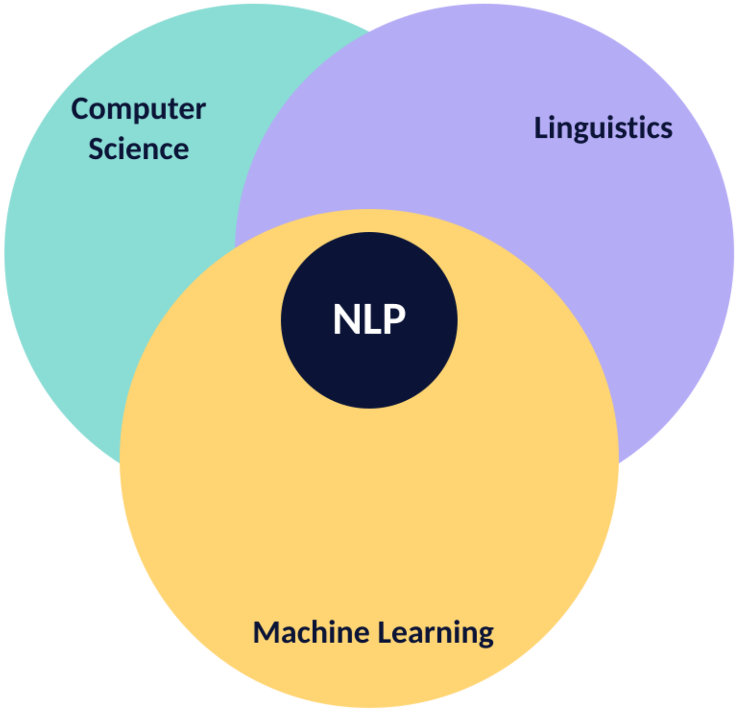
AAIS 322



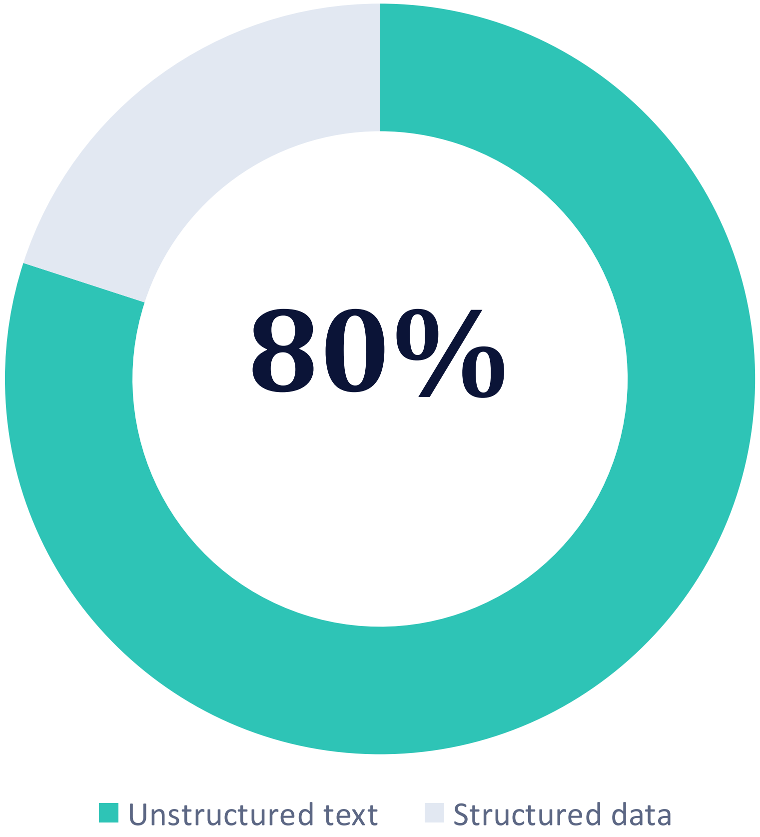
What is NLP?

- **Branch of AI & linguistics**
- **Focused on human-computer communication using natural language**
- **Converts unstructured text into structured insights**



“Teaching computers how to speak human.”

Why is NLP Important?



- **80% of world's data is text-based (emails, articles, chats)**
- **Enables search engines, chatbots, translations**
- **Drives personalization & automation**

Without NLP, the majority of the world's information is invisible to machines.

Key NLP Tasks

Text classification

Spam vs. not-spam, topic labels

Named Entity Recognition (NER)

People, places, organizations

Sentiment analysis

Positive, negative, neutral

Machine translation

English → Spanish, Chinese …

Question answering

Siri, Alexa, search snippets

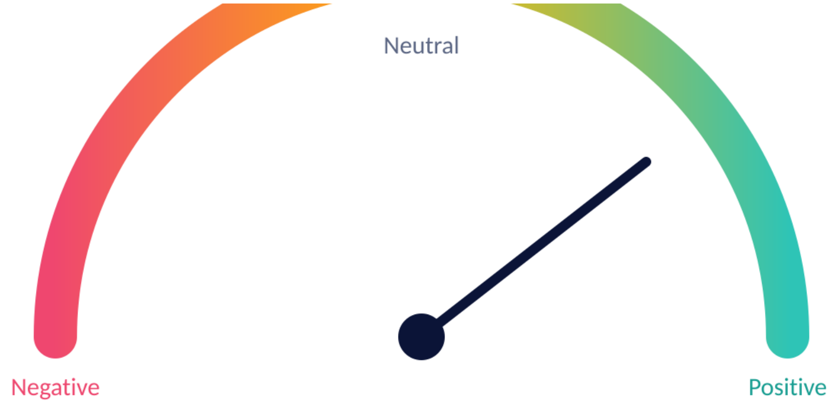
ONE PIPELINE, MANY TASKS

```
from transformers import pipeline
clf = pipeline("sentiment-analysis")
clf("NLP is incredibly useful!")
```

Example: Sentiment Analysis

"The new iPhone is amazing!" **Positive**

"The food was cold and bland." **Negative**



RULE-BASED BASELINE WITH NLTK VADER

```
import nltk; nltk.download("vader_lexicon")
from nltk.sentiment import SentimentIntensityAnalyzer

sia = SentimentIntensityAnalyzer()
sia.polarity_scores("The new iPhone is amazing!")
# {'neg': 0.0, 'neu': 0.47, 'pos': 0.53, 'compound': 0.6239}
```

Example: Chatbots

Virtual assistants

Siri, Alexa, Google Assistant

Customer service bots

Instant answers, fewer tickets

Healthcare support bots

Reliable info and reminders

AI

What's the weather tomorrow?

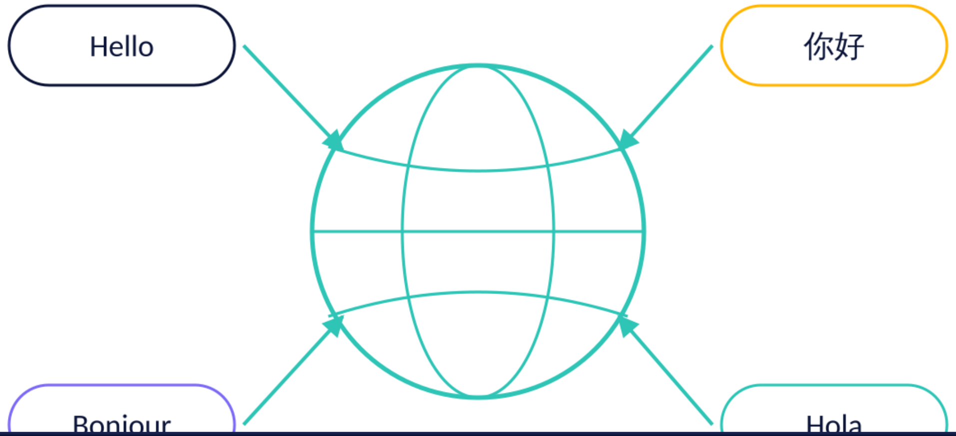
Sunny, high of 78°F downtown.
Want the hourly breakdown?

Yes, and set a reminder.

Reminder set for 7:00 AM ✓

Example: Machine Translation

- Google Translate
- Real-time multilingual communication
- Breaking down language barriers



TRANSLATION IN A FEW LINES (HUGGING FACE)

```
from transformers import pipeline

translator = pipeline("translation_en_to_fr")
translator("Natural language processing is powerful.")
# [{'translation_text': 'Le traitement du langage naturel est puissant.'}]
```

Core Components of NLP

Syntax (structure)

How words are arranged into sentences

Semantics (meaning)

What those words actually refer to

Pragmatics (context)

Tone, situation, culture — “Nice job” as praise or sarcasm

Pragmatics
context, intent, tone

Semantics
meaning of words & phrases

Syntax
structure & grammar

Traditional NLP Techniques

Raw text → Tokenize → Remove stop-words → Stem / Lemmatize → Vectorize (BoW, TF-IDF)

Tokenization

Split text into words or sub-words

Stop-word removal

Drop “the”, “and”, “of” …

Stemming & Lemmatization

“running” → “run”

Bag-of-Words (BoW)

Text as word counts

TF-IDF

Weight words by how distinctive they are

TF-IDF IN THREE LINES

```
from sklearn.feature_extraction.text \
    import TfidfVectorizer
X = TfidfVectorizer().fit_transform(docs)
```

Example: Tokenization

"I love natural language processing."

↓

"I"	"love"	"natural"	"language"	"processing"	"."

NLTK

```
import nltk
nltk.download("punkt")
from nltk.tokenize import word_tokenize

s = "I love natural language processing."
word_tokenize(s)
# ['I', 'love', 'natural', 'language',
#  'processing', '.']
```

SPACY

```
import spacy
nlp = spacy.load("en_core_web_sm")

s = "I love natural language processing."
doc = nlp(s)
[t.text for t in doc]
# ['I', 'love', 'natural', 'language',
#  'processing', '.']
```

From Rules to Machine Learning

Rule-based

1950s – 1980s

Hand-written grammars

Statistical ML

1990s – 2013

Probabilities from corpora

Deep learning

2013 – today

Embeddings & transformers

Early NLP: hand-written grammar rules

Precise, but brittle and impossible to scale

Modern NLP: statistical models + machine learning

Learn probabilities from large corpora

Deep learning revolution

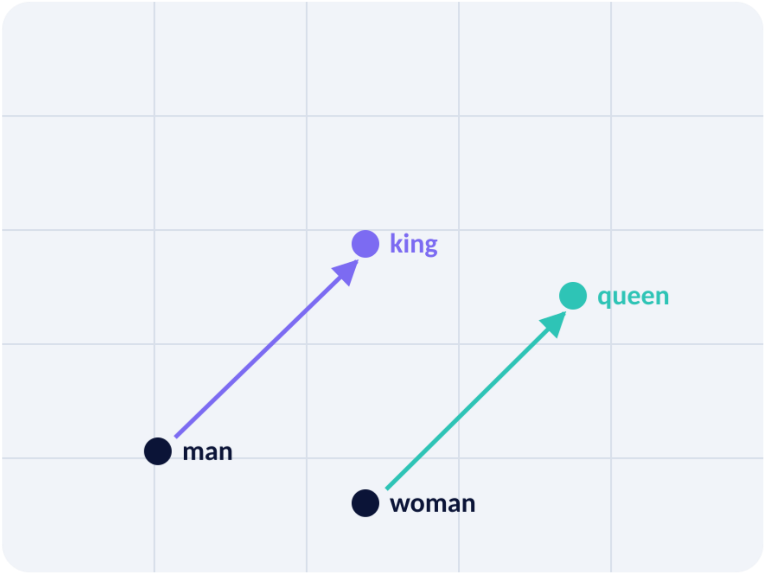
Patterns learned directly from raw text

Word Embeddings

- Word2Vec, GloVe, FastText
- Represent words as vectors
- Captures semantic meaning (“king – man + woman ≈ queen”)

TRAIN YOUR OWN WITH GENSIM

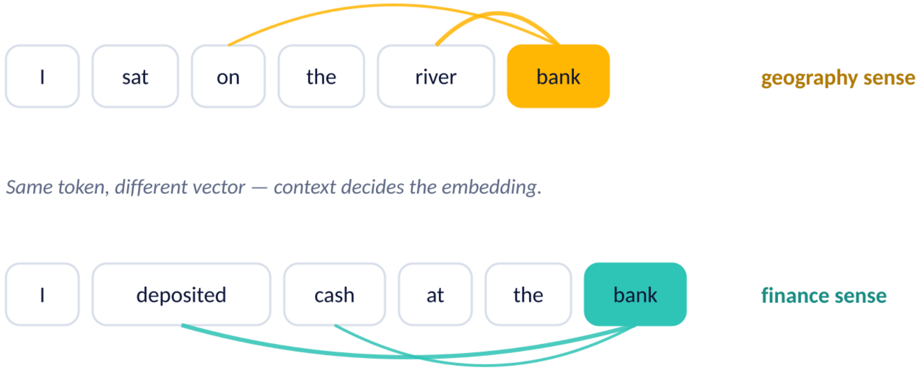
```
from gensim.models import Word2Vec

model = Word2Vec(sentences, vector_size=100,
                 window=5, min_count=2)
model.wv.most_similar(
    positive=["king", "woman"],
    negative=["man"])    # -> [('queen', 0.78), ...]
```



king − man + woman ≈ queen

Similar words end up close together in vector space.

Transformer Revolution



Same token, different vector — context decides the embedding.

- **BERT, GPT, T5**
- **Contextual embeddings**
- **State-of-the-art performance**

CONTEXTUAL EMBEDDINGS WITH BERT

```
from transformers import AutoTokenizer, AutoModel

name = "bert-base-uncased"
tok = AutoTokenizer.from_pretrained(name)
model = AutoModel.from_pretrained(name)

ids = tok("river bank", return_tensors="pt")
out = model(**ids).last_hidden_state
out.shape    # one 768-d vector per token
```

IN PRODUCTION

Real-World Applications

Healthcare

Disease prediction from clinical notes

Finance

Fraud detection and market sentiment

Legal

Contract analysis and case research

Marketing

Customer insights and personalization

Not futuristic — running today in hospitals, banks, law firms and marketing teams.

Tools & Libraries

NLTK

Classical NLP toolkit — built for teaching and research

spaCy

Modern, production-ready library — fast and opinionated

Hugging Face Transformers

Thousands of pre-trained models, one API

GET SET UP

```
pip install nltk spacy transformers scikit-learn
python -m spacy download en_core_web_sm   # small English model
```

NLTK Overview

- Great for teaching & research
- Includes tokenization, stemming, POS tagging
- Slower, not designed for production

Think of NLTK as your training wheels — excellent for learning how NLP works under the hood.

NLTK IN PRACTICE

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer

text = "The striped bats are hanging"
tokens = word_tokenize(text.lower())

# 1. drop stop-words
stop = set(stopwords.words("english"))
tokens = [t for t in tokens if t not in stop]

# 2. lemmatize + POS tag
lem = WordNetLemmatizer()
roots = [lem.lemmatize(t, "v") for t in tokens]
nltk.pos_tag(roots)
# [('striped','JJ'), ('bat','NN'), ('hang','VB')]
```

DEEP DIVE

spaCy Overview

- Industrial-strength NLP
- Optimized in Cython → very fast
- Easy integration with deep learning frameworks
- Pre-trained models

The industry choice when you need speed, accuracy and scale.

ONE PASS, MANY ANNOTATIONS

```
import spacy
nlp = spacy.load("en_core_web_sm")

doc = nlp("Apple is hiring engineers in Austin")

# named entities
for ent in doc.ents:
    print(ent.text, ent.label_)
# Apple ORG   |   Austin GPE

# POS tags and dependencies, precomputed
for tok in doc[:3]:
    print(tok.text, tok.pos_, tok.dep_)
# Apple PROPN nsubj | is AUX aux | hiring VERB

doc.vector.shape   # document embedding
```

NLTK vs spaCy

Feature	NLTK	spaCy
Purpose	Teaching, research	Production, industry
Speed	Slower	Very fast
Models	Basic	Pre-trained, deep learning ready
Community	Academic	Industry & open-source
Learning curve	Beginner-friendly	Developer-friendly

NLTK is great for learning. spaCy is great for doing.

Case Study: Twitter Sentiment

1. **Collect** — #BrandX tweets
2. **Preprocess** — clean, tokenize
3. **Vectorize** — features / embeddings
4. **Classify** — pos / neg / neutral
5. **Dashboard** — trends & alerts

- Collect tweets with hashtags (#BrandX)
- Use spaCy for preprocessing
- Classify as positive/negative/neutral

Real-time monitoring turns a bad launch into a fast response.

PREPROCESS + CLASSIFY

```python
import spacy, re
nlp = spacy.load("en_core_web_sm")

def clean(tweet):
    tweet = re.sub(r"http\S+|@\w+", "", tweet)
    doc = nlp(tweet.lower())
    keep = [t.lemma_ for t in doc
            if not t.is_stop and t.is_alpha]
    return " ".join(keep)

clf.predict(vec.transform([clean(tweet)]))
```

Case Study: Healthcare NLP

UNSTRUCTURED NOTE

```
68 y/o M w/ SOB x3 days, hx of
CHF and T2DM. Started on 40mg
furosemide. Denies chest pain.
F/u echo scheduled next week.
```

STRUCTURED DATA

AGE	68
SYMPTOM	shortness of breath
CONDITION	CHF, type 2 diabetes
DRUG	furosemide 40 mg

- **Extract symptoms from clinical notes**
- **Detect disease progression**
- **Assist in diagnosis**

Turning free text into structured data saves clinicians time and improves patient care.

REALITY CHECK

Challenges in NLP

Ambiguity of language

One phrase, many possible readings

Sarcasm & irony

“Nice job” — praise or insult?

Low-resource languages

Billions of speakers, little training data

Bias in training data

Models reproduce and amplify what they learn

These are not just technical problems — they carry ethical and social weight.

Future of NLP

More multilingual models	Dozens of languages in one system
Better handling of context & reasoning	Intent and tone, not just words
Conversational AI (ChatGPT, Claude)	Assistants that hold a thread
Ethical & fair AI systems	Systems we can actually trust

Not just smarter models — models we can trust.

Ethical Considerations

Bias in language models

Training data shapes who the model serves well — and who it fails

Misinformation generation

Fake articles and misleading reviews at scale

Privacy concerns

Medical records and private conversations are text too

Ethics is no longer optional — it is central to the field.

WRAPPING UP

Recap

NLP = AI + linguistics

Language understood at machine scale

From rules → ML → Transformers

Three eras, each a step change

Tools: NLTK, spaCy, Hugging Face

Learn, ship, and stand on pre-trained models

Real-world impact in many industries

Healthcare, finance, legal, marketing

NLP helps us make sense of human language at scale.

HANDS-ON

Run every example yourself

nlp_examples.ipynb — a companion notebook with the full, runnable version of every snippet in this deck.

1 · Setup & installation

2 · Tokenization, stop-words, lemmatization

3 · BoW and TF-IDF from scratch

4 · Sentiment analysis: VADER, sklearn, transformers

5 · Named entity recognition with spaCy

6 · NLTK vs spaCy speed benchmark

7 · Embeddings & contextual vectors

Thank you · Questions?